Technical Datasheet Input / Output Modules with Modbus TCP Protocol with Ethernet Interface

The IO module communicates via Ethernet Interface. The module is designed to allow interface with other Modbus TCP/IP compatible devices. The module acts as an input/output device between the Modbus TCP/IP Network and the Gateway.
Digital IO module is sturdy, has low power usage and is easy to use.

Port DO Module with Modbus TCP Ethernet : -

The IO modules enclosures are DIN rail Mountable. It’s a closed Enclosure with openings for connectors and LED indicators.

The design of the modules incorporates ‘**resettable Fuses**’ to safeguard against sudden High Voltage spikes. the board also has **‘reverse polarity’** protection both for input and power connector.

the communication port also has signal isolation and additional jumper for ‘end of line’120 ohms termination resistance which may or may not be used.

Specifications

General –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw Terminal Block with max 2.5 sq mm wire connection.
Dimensions	110 mm B x 110 mm L x 50 mm H
Power	Input Power – 12 – 24 VDC or 24 V AC/DC Typical – 12V DC @ 500mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Certifications	

Relay Output –

Channels	8
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 48 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10^5
Mechanical Life	1 x 10^7
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

Configuration Settings: -

Communication Speed	10/100 Base-T Ethernet compatible Interface
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Configured through web App (Details provided with Product)
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3,4,5,6,7,8

Configuration Guide – Ethernet Modbus TCP/IP IO Module:-

1. **Open a web browser** (Google Chrome, Microsoft Edge, Firefox, etc.).
2. In the browser's **address bar**, type the **default IP address:** 192.168.0.160
3. Press **Enter**. A message will appear saying: **"Please go to** http://192.168.0.160/setup**."**
4. **Copy** this link (http://192.168.0.160/setup) and **paste it** into a new browser tab.
5. Press **Enter** again to open the **Setup / Configuration Page** of the device.

To Reset the Device to Default Settings

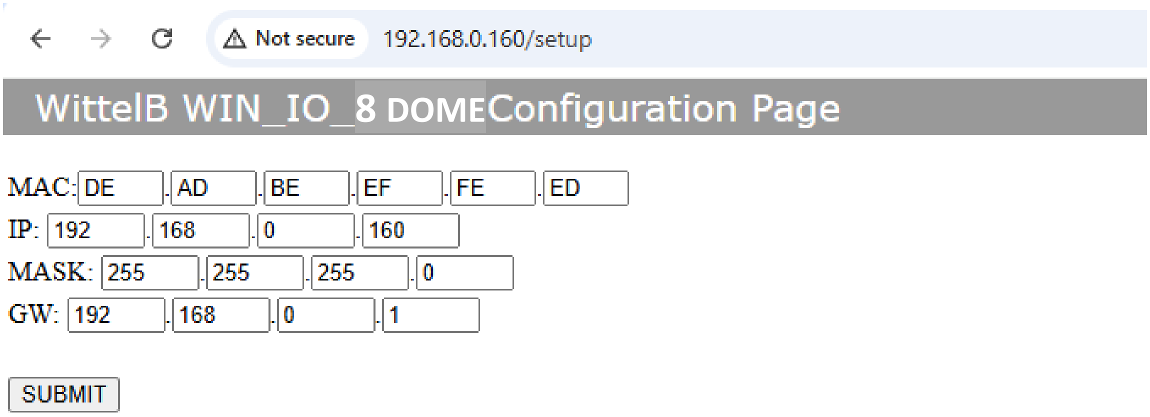
- Place the **jumper** on the designated **Reset position** on the device.
- Press the **Reset button** once.
- The module will reboot and revert to its **default IP address: 192.168.0.160.**

To Change the IP Address

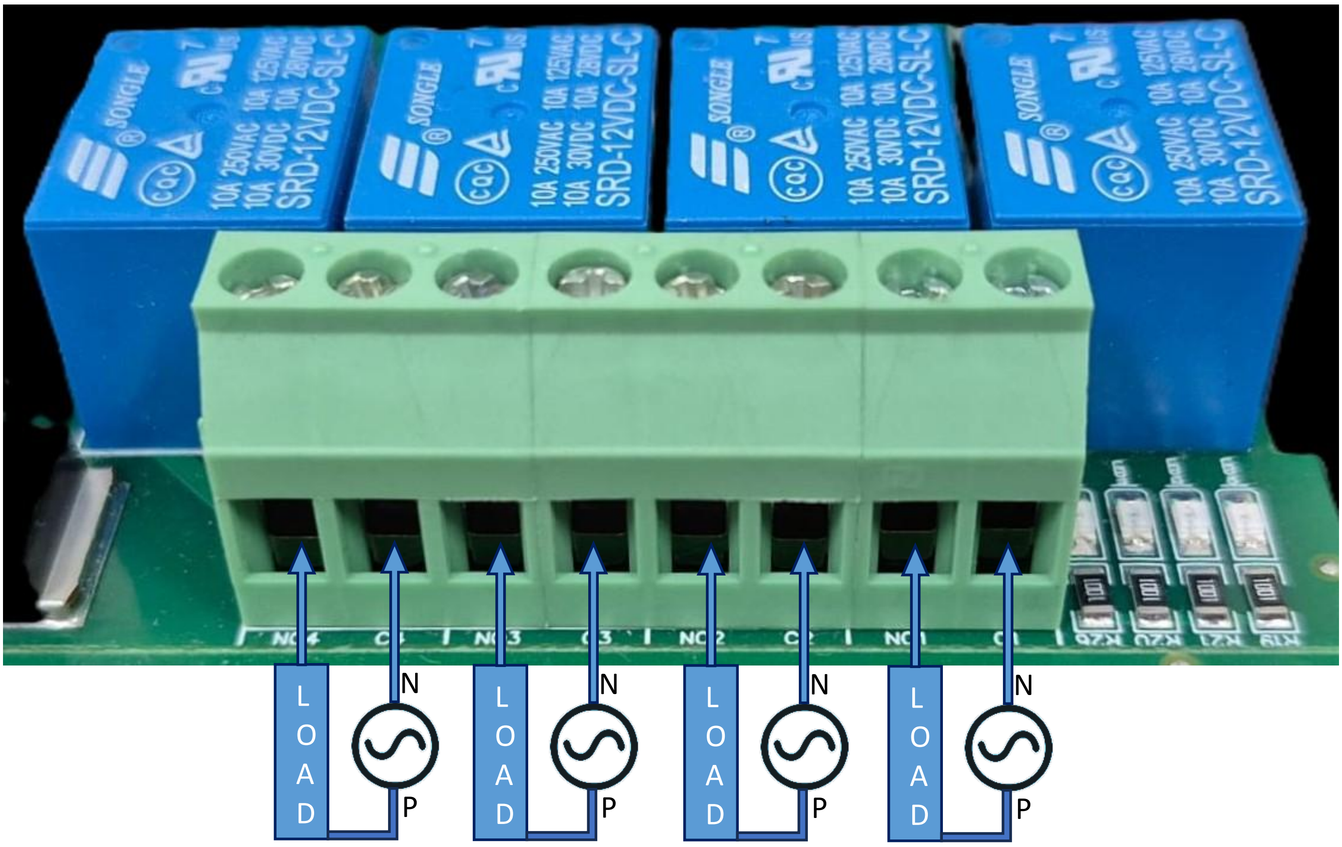
- On the **Setup / Configuration Page**, enter the **new IP address** you wish to assign.
- Click **Submit** to save the new settings.
- The device will automatically apply the **new IP address**.

Note:

- Ensure your PC's IP address is in the **same network range** (for example, **192.168.0.xxx**) when accessing the module.
- After changing the IP, you must use the **new address** to log in again.



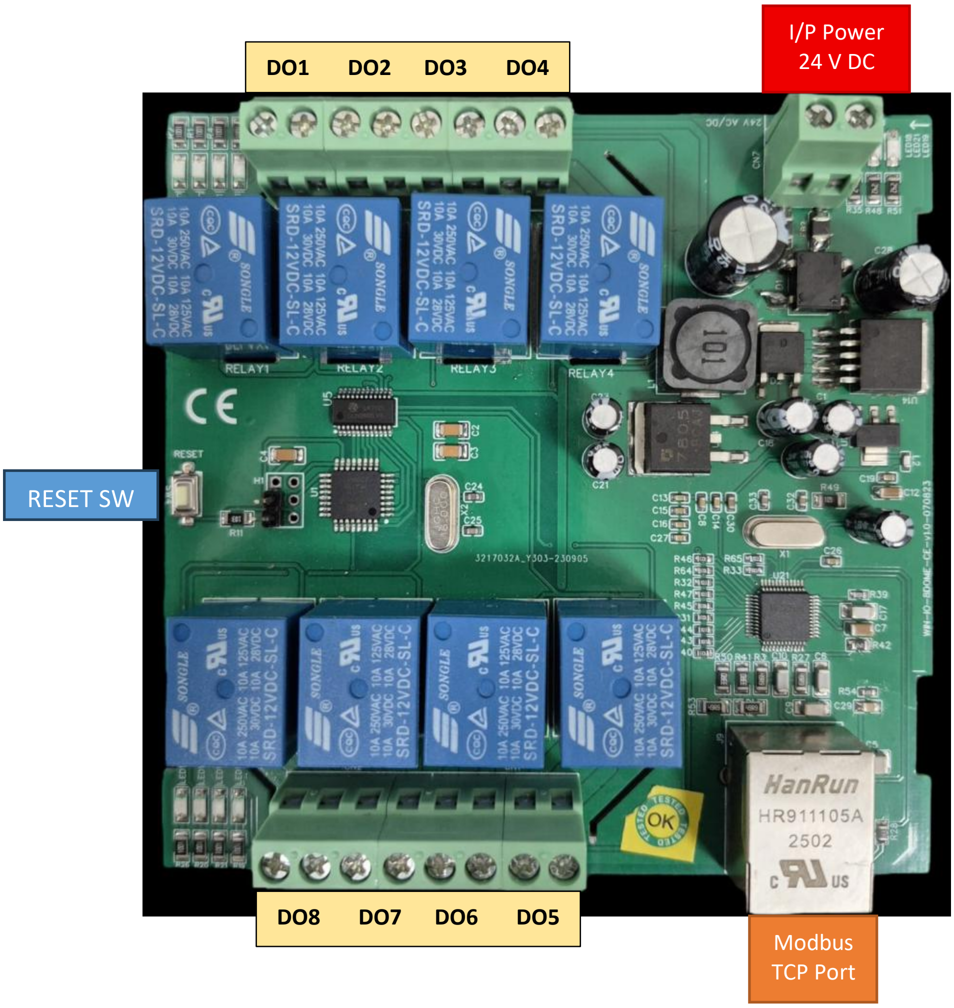
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C is internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO remains open**, and the load stays OFF.
- **Load Wiring Connections:**
  - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The **load is connected between terminal NO and Neutral (N)**.
  - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The **load is connected between terminal NO and Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

  __Note:__ Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or 10A @ 125VAC) or **10A @ 30VDC** (or 10A @ 28VDC).

Module Details –



For more details: **http://www.youtube.com/@wittelb**

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